

Revised Load Interconnection System Impact Study Report

For Queue Project Q21-1952

East Knoxville 138 kV
January 18, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1952. The Interconnection Customer proposes a 274.9 MW large load (data center) facility interconnecting at the East Knoxville 138 kV substation in the New Jersey area. The request was received on December 08, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 654 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$32,073,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: N STRASBURG - SKEENA RIVER CROSSING 138 KV
Contingency: LAMB - E KNOXVILLE 138 KV
Loading: 188.5% (Rating: 178 MVA)
DFAX: 100.0%
Pre-project loading: 34.3%
2. Facility: MILL C SUBSTATION - LAMB 138 KV
Contingency: N MESHOPPEN - KENNEDY 138 KV
Loading: 179.8% (Rating: 221 MVA)
DFAX: 66.4%
Pre-project loading: 97.1%
3. Facility: EAST KNOXVILLE - JORDAN NARROS 138 KV
Contingency: LAMB - EAST KNOXVILLE 138 KV
Loading: 162.0% (Rating: 195 MVA)
DFAX: 100.0%
Pre-project loading: 21.2%
4. Facility: E KNOXVILLE - LAMB 138 KV
Contingency: NORTH MESHOPPEN - KENNEDY 138 KV Line
Loading: 154.6% (Rating: 223 MVA)
DFAX: 92.9%
Pre-project loading: 40.3%
5. Facility: NORTH STRASBURG - JORDAN NARROWS 138 KV
Contingency: E KNOXVILLE - LAMB 138 KV L
Loading: 152.6% (Rating: 207 MVA)
DFAX: 100.0%
Pre-project loading: 19.5%

6. Facility: N MSHOPPEN - CENTRAL SAN JUAN PLANT 138 KV
Contingency: EAST KNOXVILLE SUBSTATION - LAMB SUBSTATION 138 KV
Loading: 138.3% (Rating: 187 MVA)
DFAX: 56.1%
Pre-project loading: 56.0%
7. Facility: SKEENA RIV CROSSING - CENTRAL SAN JUAN PLANT 138 KV
Contingency: EAST KNOXVILLE - LAMB SUBSTATION 138 KV
Loading: 135.3% (Rating: 192 MVA)
DFAX: 56.1%
Pre-project loading: 55.1%
8. Facility: TOMBSTONE - ASH STREET STA 223 345 KV
Contingency: MARCUS HOOK ENERGY LP - TOMSTONE 500 KV L
Loading: 128.2% (Rating: 1,413 MVA)
DFAX: 9.1%
Pre-project loading: 126.4%
9. Facility: N MESHOPPEN - KENNEDY 138 KV line
Contingency: EAST KNOXVILLE SUBSTATION - LAMB 138 KV L
Loading: 127.4% (Rating: 228 MVA)
DFAX: 56.1%
Pre-project loading: 59.8%
10. Facility: ASH ST STATION 223 - CROMBY 345 KV
Contingency: MOSER GENERATING STA - CURECANTI 345 KV
Loading: 123.2% (Rating: 1,212 MVA)
DFAX: 8.1%
Pre-project loading: 121.3%
11. Facility: COLUMBUS SOUTH SUBSTATION - DEPOSIT 138 KV
Contingency: DRISCOLL - MOUT TOM 345 KV L
Loading: 119.4% (Rating: 161 MVA)
DFAX: 8.1%
Pre-project loading: 105.6%
12. Facility: CROMBY - OWATONNA 345 KV
Contingency: CURECANTI - MOSER GENERATING STA 345 KV line
Loading: 116.7% (Rating: 1,147 MVA)
DFAX: 7.9%
Pre-project loading: 114.8%
13. Facility: DRISCOLL - JACKASS FLATS 138 KV
Contingency: ORANGE - ANSONIA 500 KV
Loading: 111.9% (Rating: 269 MVA)
DFAX: 5.3%
Pre-project loading: 106.4%
14. Facility: E HILLS - COLUMBUS S 138 KV
Contingency: MOUNT TOM - DRISCOLL 345 KV
Loading: 111.3% (Rating: 174 MVA)
DFAX: 8.1%
Pre-project loading: 98.5%
15. Facility: ONTELAUNEE ENERGY CTR 500/345kV Transformer
Contingency: TOMBSTONE - BIXBY HILL 500 KV
Loading: 110.2% (Rating: 1,080 MVA)
DFAX: 5.3%
Pre-project loading: 108.8%
16. Facility: TOMBSTONE 500/345kV Transformer
Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500 KV

- Loading: 109.1% (Rating: 1,608 MVA)
DFAX: 7.7%
Pre-project loading: 107.8%
17. Facility: CORNELIUS DELIVERY NO 1 - BERREND A C SUBSTATION 345 KV
Contingency: OWATONNA - MOSER GENERATING STA 345 KV
Loading: 107.2% (Rating: 1,223 MVA)
DFAX: 8.4%
Pre-project loading: 105.3%
18. Facility: CLEAR SPRINGS 500 kV/345 kV XFMR
Contingency: BURNS - HILLS CHAPEL 345 KV
Loading: 107.0% (Rating: 1,702 MVA)
DFAX: 11.9%
Pre-project loading: 105.1%
19. Facility: WALNUT - CLEAR SPRINGS 345 KV L
Contingency: WALLENPAUPACK 500-345 kV XFMR
Loading: 105.5% (Rating: 1,676 MVA)
DFAX: 10.0%
Pre-project loading: 103.9%
20. Facility: WALNUT CRK - SUNBURY 345 KV line
Contingency: CORONET - WESTWOOD GENERATION LLC 500 KV
Loading: 102.9% (Rating: 1,114 MVA)
DFAX: 5.5%
Pre-project loading: 101.5%
21. Facility: DEPOIT SUBSTATION - WASSON SWITCHING 138 KV
Contingency: MOUNT TOM - DRISCOLL 345 KV
Loading: 102.6% (Rating: 198 MVA)
DFAX: 8.1%
Pre-project loading: 91.4%
22. Facility: TOMBSTONE - CONNELL 345 KV
Contingency: TOMBSTONE - MARCUS HOOK ENERGY LP 500 KV
Loading: 102.5% (Rating: 2,163 MVA)
DFAX: 9.7%
Pre-project loading: 101.3%
23. Facility: WEST GALLATIN - WANUT 345 KV line
Contingency: WALLENPAUPACK 500/345 kV XFMR
Loading: 102.4% (Rating: 2,020 MVA)
DFAX: 11.1%
Pre-project loading: 100.9%
24. Facility: OWATONNA - MOSER ENERATING STA 345 KV
Contingency: LOCHBUIE - MOSER GENERATING STA 345 KV
Loading: 101.5% (Rating: 2,249 MVA)
DFAX: 10.4%
Pre-project loading: 100.2%
25. Facility: MOSER GENERATING STA - CURECANTI 345 KV
Contingency: TOMBSTONE - BIXBY HILL 500 KV line
Loading: 101.2% (Rating: 3,347 MVA)
DFAX: 15.4%
Pre-project loading: 99.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N STRASBURG - SKEENA RIVER CROSSING 138 KV	Rebuild line with higher-capacity conductor	\$2,232,000	100% (\$2,232,000)
MILL C SUBSTATION - LAMB 138 KV	Rebuild line with higher-capacity conductor	\$2,528,000	76% (\$1,932,000)
E KNOXVILLE - JORDAN NARROWS 138 KV L	Rebuild line with higher-capacity conductor	\$6,391,000	100% (\$6,391,000)
E KNOXVILLE - LAMB 138 KV	Rebuild line with higher-capacity conductor	\$656,000	100% (\$656,000)
N STRASBURG - JORDAN NARROWS 138 KV Line	Rebuild line with higher-capacity conductor	\$1,568,000	100% (\$1,568,000)
N MSHOPPEN - CENTRAL SAN JUAN PLANT 138 KV	Rebuild line with higher-capacity conductor	\$2,616,000	82% (\$2,159,000)
SKEENA RIV CROSSING - CENTRAL SAN JUAN PLANT 138 KV	Rebuild line with higher-capacity conductor	\$1,405,000	82% (\$1,160,000)
ASH ST STA 223 - TOMBSTONE 345 KV	Rebuild line with higher-capacity conductor	\$29,595,000	4% (\$1,033,000)
N MESHOPPEN - KENNEDY 138 KV	Rebuild line with higher-capacity conductor	\$1,969,000	82% (\$1,625,000)
CROMBY - ASH ST STA 223 345 KV line	Rebuild line with higher-capacity conductor	\$25,627,000	5% (\$1,237,000)
COLUMBUS SOUTH SUBSTATION - DEPOSIT 138 KV	Rebuild line with higher-capacity conductor	\$3,725,000	34% (\$1,256,000)
CROMBY - OWATONNA 345 KV	Rebuild line with higher-capacity conductor	\$18,216,000	5% (\$977,000)
DRISCOLL - JACKASS FLATS 138 KV	Rebuild line with higher-capacity conductor	\$1,405,000	53% (\$742,000)
COLUMBUS S - EAST HILLS 138 KV	Rebuild line with higher-capacity conductor	\$1,924,000	34% (\$649,000)
ONTELAUNEE ENERGY CTR 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$17,614,000	4% (\$789,000)
TOMBSTONE 500/345 kV TX	Replace transformer with higher-rated unit	\$37,885,000	3% (\$1,227,000)
CORNELIUS DELIVERY NO 1 - BERREND A C 345 KV	Reconductor line	\$13,038,000	4% (\$548,000)
CLEAR SPRINGS 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$22,140,000	4% (\$856,000)
WALNUT - CLEAR SPRINGS 345 KV line	Reconductor line	\$14,186,000	4% (\$564,000)
WALNUT CRK - SUNBURY 345 KV	Reconductor line	\$43,981,000	5% (\$2,181,000)
DEPOIT SUBSTATION - WASSON SWITCHING 138 KV	Reconductor line	\$4,092,000	34% (\$1,380,000)
TOMBSTONE - CONNELL 345 KV	Reconductor line	\$5,708,000	3% (\$151,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
WEST GALLATIN - WANUT 345 KV line	Reconductor line	\$8,938,000	4% (\$344,000)
OWATONNA - MOSER GENERATING STATION 345 KV	Reconductor line	\$4,079,000	3% (\$112,000)
MOSER GENERATING STA - CURECANTI 345 KV	Reconductor line	\$10,749,000	3% (\$304,000)
	Total allocated to project		\$32,073,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.